GenC Program

ADM DotNet Standard - Handbook

Why do we need this Academy Enablement Program?

Academy enablement program engages young talents with a comprehensive learning pathway, giving these millennials an opportunity to interact with Subject Matter Experts (SME) and understand the corporate environment and groom themselves even before they join us.

Our **Blended Learning Approach** shifts the focus from traditional teaching to active learning, empowering you to navigate your own development using our integrated tools and resources. By combining expert-led guidance with self-paced exploration, this model ensures you take full ownership of your technical growth. Get ready to embark on your own learning adventure!

Program Overview

Learning consisting of 2 Stages and an Integrated Development Project:

- **Stage 1** - Foundation in Software Development
- **Stage 2** - Application Development and Maintenance Practices
- Integrated Development Project (IDP)

Program Highlights

- **Applied Adult Learning:** A formalized journey prioritizing problem-solving and the practical application of skills over passive conceptual learning.
- **SME Mentorship:** Direct access to Business Unit (BU) experts for professional guidance, motivation, and accelerated technical growth.
- **Blended Learning & Autonomy:** Empowerment through world-class digital resources, allowing learners to drive their own progress independent of traditional instructor constraints.
- **Project-Based Learning (PBL):** Hands-on experience with the complete project lifecycle, fully integrated with Agile practices and methodologies.

Service Lines

Service lines can simply be defined as a modern organizational structure strategy for resource planning and allocation for any size of business. Typically, traditional organizational structure models are more vertically aligned -- think of an employee who has several bosses in the hierarchical ladder before being directly under the company's owner or president. Conversely, service lines follow a more horizontal continuum approach, where the company is strategically segmented into more manageable departments. The service line approach tends to focus more on the requirements of customers, which often results in noticeable increases in the customer satisfaction rate.

What is Application Development?

Application development goes through a process of planning, creating, testing, and deploying an information system, also known as the software development lifecycle. Applications are also often developed to automate some type of internal business process or processes, build a product to address common business challenges, or drive innovation.

What is Application Maintenance?

Application maintenance is the continuous updating, analyzing, modifying, and re-evaluating of your existing software applications. Application maintenance must be an ongoing task to ensure your applications are always running to the best of their abilities. Due to evolving customer expectations, the fight to survive in an existing market, and technological advancements, modifying and implementing new strategies is critical in maintaining sustainability and staying competitive. Every competitive business need to constantly enhance and manage the IT solutions that have been developed in order to stay relevant and meet the wavering needs of users. This is where application maintenance and support come into the picture.

Contrary to popular belief, application maintenance is not just about fixing defects but modifying a software product after delivery to correct faults, as well as to improve performance. Application maintenance and enhancement to existing applications begin with a thorough study of existing applications to identify areas of improvement.

Tips for Successfully Carrying Out Application Development and Maintenance

Great user experience to end customers through the development and maintenance of modern apps is a must-have. Today, applications (web or mobile) are the most cost-effective and powerful ways to reach out to a vast market and generate revenues. With millions of applications being rolled out every day, it's a good idea to keep in mind a few tips:

- Be as clear as possible as to what your requirements for your application are
- Thoroughly understand the services offered by application development companies and identify the right partner if you're using a partner
- Evaluate the various development platforms and choose the one that best fits the needs of your business

- Make sure to embed processes that focus on continuous improvements and iterations to add new features and/or fix bugs
- When developing your application, make security your top priority
- Regularly update and test your application to deliver improved and better performance, high security, and a bug-free, seamless user experience

Integrated Development Project (IDP)

What is Integrated Development Project (IDP)?

Integrated Development Project is an approach wherein the learner experiences the entire software development processes in an incremental fashion as part of the GenC Training. The IDP implementation is purely based on **Agile Software Development** methodologies and inspired from **PBL (Project-Based Learning)** which is learning while doing. It gives learners the opportunity to gain a deeper understanding of a topic through problem-solving using real-world examples and challenges.

Recommended Program Sequence

The learning journey starts with **5 days of Icebreaker sessions** followed by technical learning that contains **2 stages** along with a **Project.**

- **Stage 1** Foundation in Software Development
  - Agile Methodology
  - Software Support and Maintenance
  - Unix Commands & Shell Scripting Basics
  - GIT
  - HTML5, CSS3, JavaScript
  - Bootstrap 5
  - TypeScript
  - ANSI SQL using SQL Server
  - C# 12
  - AI Accelerate
  - **Stage 1 Assessment**
- **Stage 2** - Application Development and Maintenance Practices + **Integrated Development Project (IDP)**
  - Design Principles, UML Basics
  - Entity Framework Core 8.0
  - ASP.NET Core 8.0
  - **Interim Evaluation (Project + Technical)**
  - Building RESTful APIs with ASP.NET Core 8.0
  - Microservices Architecture
  - Application Debugging using Visual Studio Debugger
  - NUnit & Moq
  - Cognizant Flowsource 101 foundation program [101-BASICS] (ELRNG02089)
  - ITIL(PS0318_V1)-Learning
  - ITIL- Assessment
  - Windows Service
  - Python 3

- Cloud and DevOps Essentials
- GitHub Copilot
- **Final Evaluation (Project + Technical)**
- **Final HackerRank Assessment (C#, SQL + MCQ)**

Key Learning and Evaluation Components of the Program

Key Learning Components

This program follows a **blended learning model** designed to provide you with both deep conceptual understanding and the flexibility to master skills at your own pace. The journey is divided into two core pillars:

1. Trainer-Led Classroom Facilitation

The heart of our program is the interactive, live classroom environment. Throughout the training days, a dedicated expert trainer will be present to guide you through the curriculum.

- **Concept Deep-Dives:** Instead of just lectures, our trainers facilitate discussions to simplify complex technical concepts, ensuring you understand the "why" behind the "how."
- **Live Demonstrations:** You will witness real-time technical demos—ranging from cloud architecture setups to live coding sessions—providing a blueprint for your own practical work.
- **Real-Time Feedback:** The classroom serves as a safe space to ask questions, troubleshoot errors immediately, and engage in collaborative problem-solving with your peers.

2. Self-Paced Exploration via Udemy

To complement the classroom sessions, you will have access to curated **Udemy** courses. This component empowers you to take ownership of your professional development.

- **Reinforced Learning:** Use these high-quality video resources to revisit challenging topics covered in class or to explore advanced nuances of a specific technology.
- **Flexible Pacing:** While the classroom sets the rhythm, Udemy allows you to pause, rewind, and fast-forward based on your personal learning speed.
- **Continuous Upskilling:** These courses serve as a bridge between foundational training and industry-grade expertise, offering additional practice exercises, quizzes, and downloadable resources to solidify your technical toolkit.

RAG as PHS (Performance Health Status)

The program continuously evaluates your ability to apply self-learnt skills to solve real-time business problems. The following two key components are distributed across the learning journey to facilitate **Continuous Evaluation**.

1. Interim Evaluation

Occurring at the halfway point of the program, this stage assesses foundational knowledge and initial application.

- **Format:** Video interview on the learning platform conducted by a **Technical Subject Matter Expert (SME)** from the Business Unit.
- **Scope:** Assessment of core skills covered in the first half of the training.
- **Project Review:** Evaluation of your **Integrated Development Project (IDP)** progression to gauge early-stage proficiency and technical logic.

2. Final Evaluation

A comprehensive assessment at the conclusion of the program to validate end-to-end expertise.

- **Format:** Comprehensive video interview with a **Technical SME**.
- **Scope:** Full-spectrum evaluation of all skills and technologies covered throughout the entire training journey.
- **Project Defense:** Final project evaluation of the **IDP** to assess your capability to deliver a business-ready solution and demonstrate overall technical mastery.

[! IMPORTANT]

The evaluation components above are the primary drivers of your **Performance Health Status (PHS)**.

Enhancing Your Expertise

While the evaluations determine your PHS, the following **Additional Learning Components** are integrated into the journey to help you sharpen your skills and prepare for assessments:

- **Hands-On:** Practical lab exercises for technical muscle memory.
- **Quizzes:** Rapid knowledge checks on key concepts.
- **CCs (Code Challenges):** Scenario-based problem-solving.
- **ICTs (Integrated Capability Tests):** Periodic mock assessments to track your growth.

Stage 1 - Foundation in Software Development

Overview

Stage 1 deals with foundational technology skills that help GenCs to get start with their software engineering career. We provide unique learning experience to learners by including diversified learning content and learning methodologies that are based on adult learning principles.

As part of Stage 1 of your training, the following skills will be covered.

- Agile Methodology

- Software Support and Maintenance
- Unix Commands & Shell Scripting Basics
- GIT
- HTML5, CSS3, JavaScript
- Bootstrap 5
- TypeScript
- ANSI SQL Using SQL Server
- C# 12

How and From Where to Learn?

- **Udemy learnings are recommended** to understand the fundamental concepts. In addition to this, you can also learn from any other sources as they are mentioned in this handbook.

Stage 1 -> Course 1 -> Agile Methodology

Course Overview

In the **Course 1** of the **Stage 1**, learners will be introduced to the basics of **Agile methodology**. Agile is an approach to project management and software development that emphasizes flexibility, collaboration, and customer satisfaction. It involves adaptive planning, iterative development, early delivery, and continuous improvement. Agile methodologies, like Scrum and Kanban, focus on delivering value to the customer and responding to change effectively.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the principles and values of Agile methodology.
- Describe the benefits of using Agile in software development.
- Explain the differences between Agile and traditional project management approaches.
- Identify the key roles and responsibilities in Agile teams.
- Describe the iterative and incremental nature of Agile development.
- Explain the importance of customer collaboration and feedback in Agile.
- Describe common Agile practices, such as user stories, sprints, and retrospectives.
- Identify common Agile frameworks, such as Scrum, Kanban, and Extreme Programming (XP).
- Explain how Agile principles can be applied in different project environments.

Day 1

Agile Methodology

Key Topics: Introduction to Agile, Agile Manifesto, Scrum Framework, Agile Estimation and Planning, Agile User Stories, Agile Metrics and Reporting

Continuous Learning: Technical Enablement

Agile Crash Course: Agile Project Management; Agile Delivery
- Learn All Sections in this Udemy course.

Stage 1 -> Course 2 -> Software Support and Maintenance

Course Overview

Course 2 of **Stage 1** provides a comprehensive understanding of software development and maintenance processes, focusing on the Software Development Life Cycle (SDLC) phases and Agile methodology. It explores the critical need for software maintenance, challenges encountered during the maintenance phase, and the various categories of maintenance activities. Additionally, learners will gain insights into software reverse engineering and its significance, along with best practices for providing effective software support. The course is designed to equip learners with the necessary skills to ensure the efficiency and longevity of software systems in dynamic environments.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain the stages of the Software Development Life Cycle and their role in delivering robust software solutions.
- Demonstrate knowledge of Agile principles and practices to enhance flexibility and collaboration in software development and maintenance.
- Articulate the importance of software maintenance in ensuring system performance and alignment with evolving requirements.
- Analyze common challenges in software maintenance and propose strategies to address them effectively.
- Differentiate between various types of software maintenance, including corrective, adaptive, perfective, and preventive maintenance.
- Describe the purpose and process of software reverse engineering in understanding legacy systems and facilitating updates.
- Develop strategies for providing efficient and proactive software support to meet user and business needs.

Day 2 - Forenoon

Software Support and Maintenance

Key Topics: SDLC vs Agile, Software maintenance and support - overview

Continuous Learning: Technical Enablement

Software Engineering 101: Plan and Execute Better Software

- Learn the sections listed below in this Udemy course and take up the Quizzes in each section in order to check your understanding about the subject.
    - **Section 2:** Software Lifecycle
    - **Section 3:** Requirements and Specifications
    - **Section 4:** Design: Architecture
    - **Section 5:** Design: Modularity
    - **Section 6:** Implementation and Deployment
    - **Section 7:** Testing
    - **Section 8:** Software Development Models

Additional Learning

Learn about Software Maintenance from the following:

- Overview of Software Maintenance

Stage 1 -> Course 3 -> Unix Commands & Shell Scripting Basics

Course Overview

Course 3 of **Stage 1** is designed to provide learners with a comprehensive introduction to Unix and Unix-based systems. It covers essential skills required for effective file and directory management, mastery of file permissions, and the use of basic utilities. Learners will explore powerful tools like pipes, filters, and text processing commands to manipulate data efficiently. The course also delves into process management and network communication utilities, laying a strong foundation for Unix system usage. Additionally, participants will gain hands-on experience with Unix shell scripting, from writing their first script to understanding scripting fundamentals and incorporating error-handling techniques. This course equips learners with the skills needed to automate tasks, streamline workflows, and enhance their productivity in a Unix environment.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the Unix operating system and its core functionalities.
- Perform effective file and directory management using Unix commands.
- Configure and modify file permissions and access modes for secure operations.
- Utilize basic utilities, pipes, and filters to process and manipulate data efficiently.
- Apply text processing commands to manage and analyze text-based data.
- Manage processes and monitor system performance using Unix process management tools.
- Utilize network communication utilities for system interaction and data transfer.
- Develop a foundational understanding of Unix shell scripting, including scripting syntax and structure.
- Write and execute basic shell scripts to automate routine tasks.
- Implement error-handling techniques in shell scripts to ensure robust script execution.

Day 2 - Afternoon

Unix Commands & Shell Scripting Basics

Key Topics: Introduction, File Management, Directory Management, File Permission / Access Modes, Basic Utilities, Pipes and Filters, Process Management, Network Communication Utilities

Continuous Learning: Technical Enablement

The Linux Command Line Bootcamp: Beginner To Power User

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 2:** Introduction
    - **Section 3:** Command Basics
    - **Section 4:** Getting Help
    - **Section 5:** Navigation
    - **Section 6:** Creating Files & Folders
    - **Section 8:** Deleting, Copying, & Moving
    - **Section 10:** Working With Files
    - **Section 11:** Redirection
    - **Section 12:** Piping
    - **Section 14:** Finding Things
    - **Section 15:** Grep
    - **Section 16:** Permissions Basics
    - **Section 17:** Altering Permissions
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Display the days of the year 2016
- Calculator
- Display the hidden files or directories
- Copy a File using relative path
- Move a File to mammals using relative path
- Change File Permission adding write permission to group
- Change File permission by removing write permission as specified
- Display the lines of the file that do not contain the string
- List all the file-names starting with "t" or "s"
- Count the Files in a directory

Additional Learning

Learn about Process Management and Network Communication Utilities from the following

- Linux/Unix Process Management
- Network Communication Utilities

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on Unix and Shell Scripting.

- Introduction to Unix
  - Pre-Quiz
  - Post-Quiz
- File System
  - Pre-Quiz
  - Post-Quiz
- Filters
  - Pre-Quiz
  - Post-Quiz
- Test Your Understanding - Introduction to Unix
- Test Your Understanding - File System
- Test Your Understanding – Filters

Day 3

Unix Commands & Shell Scripting Basics

Key Topics: Introduction to Shell Scripting, Basic Shell Scripting Concepts, Control Structures, Command-Line Arguments, Functions, Text Processing, Error Handling and Debugging

Continuous Learning: Technical Enablement

Shell Scripting: Discover How to Automate Command Line Tasks

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 2:** Shell Scripting, Succinctly
  - **Section 5:** Shell Script Checklist and Template
  - **Section 7:** Case Statements
  - **Section 9:** While Loops
  - **Section 3:** Exit Statuses and Return Codes
  - **Section 4:** Functions
  - **Section 8:** Logging
  - **Section 10:** Debugging
  - **Section 11:** Data Manipulation and Text Transformation with Sed
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Permission Change
- Factorial of a number

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on Unix and Shell Scripting.

- Vi Editor
  - Pre-Quiz
  - Post-Quiz
- Bourne Shell
  - Pre-Quiz
  - Post-Quiz
- Test Your Understanding - Vi Editor
- Test Your Understanding - Bourne Shell

Stage 1 -> Course 4 -> GIT

Course Overview

Course 4 of **Stage 1** provides a comprehensive understanding of Git, a powerful version control system widely used in software development. Participants will learn the foundational concepts and advanced techniques necessary to effectively manage and collaborate on projects using Git. The course is structured to guide learners from working locally with Git to mastering collaborative workflows, including branching, merging, and rebasing. By the end of this course, participants will be well-equipped to handle real-world version control challenges and streamline their development processes.

Learning Objectives

After completing this course, GenCs will be able to:

- Grasp fundamental Git concepts, including repositories, versioning, and the role of Git in software development workflows.
- Set up and initialize local Git repositories.
- Track changes and manage versions using essential Git commands like add, commit, and log.
- Navigate through commit history and revert changes when necessary.
- Connect local repositories to remote services like GitHub or GitLab.
- Push and pull changes to and from remote repositories.
- Understand and resolve common conflicts during collaborative workflows.
- Create and manage branches for feature development and bug fixes.

- Merge changes effectively to maintain a stable codebase.
- Rebase commits to keep a clean and linear project history.

Day 4

GIT

Key Topics: Introduction, Working Locally with Git, Working Remotely with Git, Branching, Merging, and Rebasing with Git

Continuous Learning: Technical Enablement

Git Complete: The definitive, step-by-step guide to Git

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Introduction
    - **Section 2:** Installation
    - **Section 3:** Git Quick Start
    - **Section 6:** Basic Git Commands
    - **Section 8:** Comparisons
    - **Section 9:** Branching and Merging
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Git Config
- Clone Repo
- Add, Commit And Push
- Pull And Merge
- Merge - Resolve Conflict
- Git Tags

Stage 1 -> Course 5 -> HTML5, CSS3, JavaScript

Course Overview

Course 5 of **Stage 1** focuses on foundational Web UI technologies **- HTML5, CSS3,** and **JavaScript** - which are essential for creating modern, responsive, and interactive user interfaces for web applications. HTML5 defines the structure of web pages, CSS3 enables advanced styling and layout customization, and JavaScript brings interactivity and dynamic behavior, empowering developers to build visually appealing and user-friendly web experiences.

Learning Objectives

After completing this course, GenCs will be able to:

- Define HTML and common terminology related to HTML, recognize correct HTML syntax, and write a brief error-free HTML code.
- Apply style to an existing/new web page as per the requirement using CSS3.
- Write and employ JavaScript code to solve practical web design problems.

Day 5

HTML5, CSS3, JavaScript

Key Topics: HTML5 - Introduction, Getting Started, Elements & Attributes, Navigation, Events, Web Forms 2.0, Web Storage, Web SQL Database, Geo location; CSS3 - Introduction, Selectors, Styling, Box Model, Advanced Concepts

Continuous Learning: Technical Enablement

The HTML & CSS Bootcamp 2024 Edition

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Introduction
    - **Section 2:** HTML Basics
    - **Section 3:** More HTML
    - **Section 4:** Working With Forms
    - **Section 5:** Other Elements
    - **Section 6:** CSS Basics
    - **Section 7:** The World of CSS Colors
    - **Section 8:** Styling Text
    - **Section 9:** More Text Styling
    - **Section 10:** Selectors Pt. 1
    - **Section 11:** The Box Model
    - **Section 15:** CSS Units
    - **Section 17:** Backgrounds, Gradients, & Filters
    - **Section 18:** Positioning
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Newberry Library
- Jasper Sports Academy
- WhatsApp Chat
- Calculator
- 4*4 Sudoku

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level in HTML5 and CSS3.

- HTML 5, CSS3

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac for checking your skill level in HTML5 and CSS3. You have to secure 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - HTML5 and CSS3

Day 6

HTML5, CSS3, JavaScript

Key Topics: Getting Started, Fundamentals, Operators, Control Flow, Objects, Arrays

Continuous Learning: Technical Enablement

Javascript basics for beginners

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Getting Started
  - **Section 2:** Basics
  - **Section 3:** Operators
  - **Section 4:** Control flow
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Schedule Upcoming Match
- Event Reservation Portal
- Display Student Details

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac to check your knowledge level in JavaScript.

- Java Script

Additional Learning

Go through the Udemy course to understand the usage of Chrome Developer Tools, a comprehensive toolkit for developers built directly into the Chrome browser. These tools allow you to edit web pages in real-time, diagnose problems more quickly, and build better websites faster.

Devtools Pro: The Basics of Chrome Developer Tools

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **All** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 7 - Forenoon

HTML5, CSS3, JavaScript

Key Topics: JavaScript Deep Dive

Continuous Learning: Technical Enablement

Javascript basics for beginners

- Learn the sections listed below in this Udemy course and complete the corresponding hands-on coding given below.
  - **Section 5:** Objects
  - **Section 6:** Arrays
  - **Section 7:** Functions
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-on

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Counting Game
- SignUp Form validation
- Fitness Centre
- Product Code Validation

Code Challenge (For Practice Only)

Attempt the following Code Challenges through the Learning Path at Tekstac for checking your skill level on JavaScript. There will be only 3 attempts, and you have to secure 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - JavaScript

Stage 1 -> Course 6 -> Bootstrap 5

Course Overview

Course 6 of **Stage 1** provides a comprehensive introduction to **Bootstrap 5**, the world's most popular front-end framework for building responsive, mobile-first websites. GenCs will explore the fundamental concepts and tools that Bootstrap 5 offers, empowering them to design and develop modern, dynamic web interfaces efficiently. From understanding the layout and grid system to leveraging essential components, utilities, and helpers, this course will equip them with the skills to create visually appealing and user-friendly web pages. Additionally, GenCs will learn how to customize and extend Bootstrap to meet specific project requirements while mastering responsive design principles.

Learning Objectives

After completing this course, GenCs will be able to:

- Describe the purpose and features of Bootstrap 5 as a front-end framework.
- Design responsive web layouts using the Bootstrap grid system.
- Implement advanced layout techniques for complex web pages.
- Incorporate common UI elements like navigation bars, modals, buttons, and cards to enhance interactivity and usability.
- Use Bootstrap utilities and helper classes to streamline the styling and functionality of web pages.
- Develop web pages that adapt seamlessly to different devices and screen sizes.
- Leverage responsive breakpoints effectively for design adjustments.
- Modify default themes and styles to align with specific design requirements.
- Integrate Bootstrap with custom CSS and JavaScript to extend its functionality.

Day 7 - Afternoon

Bootstrap 5

Key Topics: Introduction to Bootstrap 5, Layout and Grid System

Continuous Learning: Technical Enablement

The Ultimate Bootstrap Guide - Bootstrap 5 from Scratch

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Getting Started with Bootstrap 5
  - **Section 2:** Layouts in Bootstrap 5
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Ohm's Law

Day 8 - Forenoon

Bootstrap 5

Key Topics: Essential Components, Utilities and Helpers, Responsive Design, Customization and Extensions

Continuous Learning: Technical Enablement

The Ultimate Bootstrap Guide - Bootstrap 5 from Scratch

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 5:** Components in Bootstrap 5
  - **Section 6:** Helpers in Bootstrap 5
  - **Section 7:** Utilities in Bootstrap 5
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learning References

- Responsive Web Design - Media Queries
- Responsive Web Design - Images

- Bootstrap 5 Navbar Responsive behaviors Toggler
- How to theme, customize, and extend Bootstrap with Sass

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Sticky Notes
- Portfolio
- Library Home Page
- Hi-Tech Digital world

Stage 1 -> Course 7 -> TypeScript

Course Overview

Course 7 of **Stage 1** provides a comprehensive introduction to **TypeScript**, a typed superset of JavaScript. Designed to bridge the gap between dynamic scripting and structured engineering, the program moves from basic syntax and configuration to advanced object-oriented patterns. Learners will master how to leverage TypeScript's compiler to catch errors early, write self-documenting code, and implement robust interfaces and classes for scalable application development.

Learning Objectives

After completing this course, GenCs will be able to:

- Install the TypeScript compiler and configure tsconfig.json for optimized development workflows.
- Apply primitive, special, and collection types (Arrays and Tuples) to ensure data integrity.
- Define precise parameter and return types for standard and arrow functions, including optional and default values.
- Use **Interfaces** and **Object Types** to enforce strict contracts for complex data structures.
- Implement classes with access modifiers (public, private, protected) and utilize inheritance.
- Utilize **Union** and **Literal types** alongside type narrowing to handle multi-state logic and dynamic data securely.
- Balance explicit type annotations with TypeScript's inference engine and safely override types using assertions.

Day 8 - Afternoon

TypeScript

Key Topics: Introduction to TypeScript, Basic Types, Type Annotations and Inference, Functions

Continuous Learning: Technical Enablement

Understanding TypeScript

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Getting Started
    - **Section 2:** TypeScript Basics & Basic Types
    - **Section 3:** The TypeScript Compiler (and its Configuration)
    - **Section 4:** TypeScript Essentials Demo Project
    - **Section 5:** Next-generation JavaScript & TypeScript
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 9 - Forenoon

TypeScript

Key Topics: Interfaces and Object Types, Classes and Access Modifiers, Union and Literal Types

Continuous Learning: Technical Enablement

Understanding TypeScript

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 6:** Classes & Interfaces
    - **Section 7:** Advanced Types
    - **Section 8:** Generic Types
    - **Section 9:** Classes & Generics - Demo Project
    - **Section 10:** Deriving Types From Types
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Course Overview

In **Course 7** of **Stage 1**, you will focus on **ANSI SQL programming using SQL Server**, a fundamental skill for database management. This module provides an in-depth exploration of Microsoft SQL Server, a powerful relational database management system. It is designed to equip learners with both foundational and advanced skills in SQL Server, covering everything from basic data types and queries to complex operations like joins, subqueries, stored procedures, functions, triggers, cursors, and exception handling. Through hands-on practice and real-world scenarios, learners will gain the ability to design, query, and manage SQL Server databases effectively.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand and apply ANSI SQL fundamentals in the context of SQL Server.
- Work with SQL Server instances and navigate SQL Server Management Studio (SSMS).
- Apply constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, and CHECK to enforce data integrity.
- Practice retrieving and manipulating data using the SELECT statement in SQL Server.
- Apply filtering, sorting, and grouping techniques to query data effectively.
- Gain an understanding of join operations and subqueries for combining data from multiple tables.
- Use set operations like UNION, INTERSECT, and EXCEPT to combine query results.
- Use GROUP BY, HAVING, GROUPING SETS, CUBE, and ROLLUP to aggregate and analyze data.
- Create, modify, execute, and manage stored procedures and user-defined functions.
- Understand the benefits and types of stored procedures and functions in SQL Server.
- Understand the purpose and types of triggers (AFTER, INSTEAD OF, LOGON) and how to create and manage them.
- Explore the lifecycle, types, and limitations of cursors, and identify alternatives.
- Handle errors using TRY...CATCH, THROW, and RAISERROR.
- Differentiate between system-defined and user-defined exceptions and retrieve detailed error information.

Day 9 - Afternoon, 10

ANSI SQL Using SQL Server

Key Topics: Basics, Data Types, Constraints, Querying data, Sorting data, Filtering data, Grouping data, Expressions

Continuous Learning: Technical Enablement

70-461, 761: Querying Microsoft SQL Server with Transact-SQL

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 2: Session 1 -** Starting SQL Server
    - **Section 3: Session 1 -** Creating tables - First pass
    - **Section 4: Session 1 -** Number types and functions
    - **Section 5: Session 1 -** String data types and functions
    - **Section 9: Session 2 -** Creating and Querying part of a table
    - **Section 10: Session 2 -** Summarising and ordering data
    - **Section 28: Section 5 - Objective 9b:** Grouping sets
    - **Section 34: Session 6 - Objective 7b:** PIVOTing and UNPIVOTing
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the solution. Write the query yourself.

- Flight Booking Details – DDL
- Alter Flight Details – DDL
- Doctor and Updated Doctor – Merge
- Flight Count By Source – DRL
- Car Sales – Rank
- Bill Count Based On Status - Pivot
- Boat Details Based On Seating Capacity
- Sailor Details Based On Condition
- Boat Details Based On The Type And Seating Capacity
- Weekend Flights – DRL
- Time Change – DRL
- Highest And Lowest Experience
- Shift Based Ride Info
- Display Boat Type

Learning References

- Sort and filter results in T-SQL
- Summarize data with GROUP BY
- Filter groups with HAVING

Day 11, 12, 13 - Forenoon

ANSI SQL Using SQL Server

Key Topics: Views, Joining tables, Subquery, Set Operators, Stored Procedures, User-Defined Functions, Triggers and Cursors, Exception Handling, Transaction Control in SQL Server

70-461, 761: Querying Microsoft SQL Server with Transact-SQL

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 11: Session 2 -** Adding a second table
  - **Section 32: Session 6** - Objective 7: Sub-queries
  - **Section 12: Session 2 -** Find missing data, and delete and update data
  - **Section 16: Session 3** - Objectives 2 and 3: Views
  - **Section 20: Session 4 - Objective 13:** Combine datasets
  - **Section 28: Section 5 - Objective 9b:** Grouping sets
  - **Section 21: Session 4** - Objective 11 - Create and alter stored procedures (simple statements
  - **Section 36: Session 6** - Objective 14: Functions
  - **Section 22: Session 4** - Objective 18a - Implement try/catch/throw
  - **Section 17: Session 3** - Objective 5: Create and alter DML triggers
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the solution. Write the query yourself.

- Passenger Info - Transaction Control
- Booking Amount - Procedure
- Passenger Info - Procedure
- Booking - Procedure
- Update Flight Name - Function
- Booking Information - Function
- Passenger Info - Exception Handling
- Delete Flight Details- Exception Handling
- March Booking - Cursor
- Insert - Trigger
- Update Flight Date - Trigger

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on Database concepts.

- SQLServer

Code Challenges (For Practice Only)

Attempt the following Code Challenges through the Learning Path at Tekstac for checking your skill level on SQL Programming. You must secure 70% to clear this challenge.

Do not copy paste the solution. Write the query yourself.

- Assess-Type-1: Code Challenge - SQL Server

Stage 1 -> Course 8 -> C# 12

Course Overview

Course 8 of **Stage 1** provides a comprehensive introduction to modern C# programming using C# 12 and .NET 8. It is designed to equip learners with the foundational and intermediate skills required to build robust, secure, and scalable applications. The course begins with an overview of the .NET ecosystem and development environment setup, then progresses through core programming constructs, object-oriented principles, advanced language features, asynchronous programming, file handling, multithreading, debugging, secure coding practices, and database connectivity using ADO.NET. Emphasis is placed on new features introduced in C# 12 and performance improvements in .NET 8, ensuring learners are up to date with the latest advancements in the platform.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the .NET Ecosystem and set up the development environment using .NET SDK and Visual Studio/VS Code.
- Write and execute basic C# programs, using variables, constants, data types, and type inference.
- Apply control flow constructs such as conditional and iterative statements, including enhancements in C# 12.
- Define and invoke methods with various parameter types and understand method overloading and local functions.
- Implement object-oriented programming concepts including classes, objects, constructors, access modifiers, and inheritance.
- Utilize modern C# features such as primary constructors, records, init-only setters, and pattern matching.
- Handle exceptions effectively using try-catch-finally blocks, custom exceptions, and filters.
- Work with null safety features including null-coalescing, null-conditional operators, and the required modifier.
- Manipulate collections such as lists, dictionaries, queues, and stacks, and perform LINQ queries.
- Use tuples and advanced pattern matching techniques including list patterns and switch expressions.
- Implement asynchronous programming using tasks, async/await, and understand improvements in ValueTask.
- Perform file I/O operations and JSON serialization using System.Text.Json.
- Develop multi-threaded applications, manage concurrency, and use synchronization primitives.

- Debug and trace applications using built-in tools and logging frameworks like System.Diagnostics and log4net.
- Apply secure coding practices to prevent common vulnerabilities such as XSS, CSRF, and SQL injection.
- Connect to databases using ADO.NET and perform data retrieval and manipulation using its core classes.

Day 13 - Afternoon

C# 12

Key Topics: Introduction to C# and .NET 8, C# Syntax and Basic Constructs, Control Flow Statements

Continuous Learning: Technical Enablement

C# 13 - Ultimate Guide - Beginner to Advanced | Master class

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Course Fundamentals (Theory)
    - **Section 2:** Language Basics (Practical Starts Here)
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Variables
- User Input Handling
- Sky Diving Restrictions
- Loan Eligibility
- Latecomer's Detention
- Total Finding

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on looping concept in C#.

- C# - Looping

Day 14

C# 12

Key Topics: Functions and Methods, Introduction to OOP Concepts

Continuous Learning: Technical Enablement

C# 13 - Ultimate Guide - Beginner to Advanced | Master class

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 4:** C# Object Oriented Programming - Basics
  - **Section 5:** Fields
  - **Section 6:** Methods
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Case Converter
- Inserting Word
- Discount Offer
- ConTv Company

Day 15

C# 12

Key Topics: Encapsulation and Properties, Inheritance and Polymorphism

Continuous Learning: Technical Enablement

C# 13 - Ultimate Guide - Beginner to Advanced | Master class

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 10:** Inheritance, Hiding, Overriding
  - **Section 11:** Abstract classes and Interfaces
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Machine Masters
- ROI Calculator
- Area Calculation
- Total Price

Day 16

C# 12

Key Topics: Records and Structs, Exception Handling, Nullable Types and Null Safety

Continuous Learning: Technical Enablement

C# 13 - Ultimate Guide - Beginner to Advanced | Master class

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 14:** Structures
  - **Section 17:** Handling Null
  - **Section 28:** Exception Handling
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 17

C# 12

Key Topics: Collections and LINQ

Continuous Learning: Technical Enablement

C# 13 - Ultimate Guide - Beginner to Advanced | Master class

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 16:** Generics
  - **Section 21:** Arrays
  - **Section 22:** Collections
  - **Section 2**5: LINQ
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- The Rank Finder
- Maximum Points
- Tasty Byte Beverages
- Case File Management
- Course Details

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac for checking your skill level on C# programming. You must secure 70% to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - C# - Group 1

Day 18

C# 12

Key Topics: Tuples and Pattern Matching Enhancements, Asynchronous Programming with Async/Await, File Handling and I/O Operations

Continuous Learning: Technical Enablement

C# 13 - Ultimate Guide - Beginner to Advanced | Master class

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 18:** Extension Methods and Pattern Matching
  - **Section 24:** Tuples
  - **Section 27:** IO, Serialization, Encoding
  - **Section 33:** Asynchronous Programming
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Words Finding

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on Lambda Expressions.

- C# - Lambda Expressions

Day 19

C# 12

Key Topics: Multi-Threading, Debugging and Tracing, Secure Coding

Continuous Learning: Technical Enablement

C# 13 - Ultimate Guide - Beginner to Advanced | Master class

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 31:** Threading
    - **Section 32:** Tasks
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 20, 21 - Forenoon

C# 12

Key Topics: ADO.NET Fundamentals

Continuous Learning: Technical Enablement

- Download and go through the PDF document ADO.Net_Learning.pdf which is accessible via **ADO.NET Learning** under Learning content section. Refer page numbers 1 to 21, 25 to 28

Hands-On

Complete the following hands-on exercises in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- ADO Update
- ADO Delete

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac for checking your skill level on C# programming and ADO.NET. You have to secure 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - C# - Group 2

ICT (Integrated Capability Test) (For Practice Only)

Take up the following extended integrated practice task to check your skill level after completing the Stage 1 of your training. Unlike Code Challenge, the coverage of this practice will be C#, SQL Server and ADO.NET. You must score a minimum 70% to complete this activity successfully.

- Dotnet Assess-Type-2: Integrated Capability Test (ICT)

Mock Assessment

- Ensure that you take the mock assessment before the actual one to become familiar with its pattern. You can attend the mock qualifier multiple times.

Day 21 - Afternoon

AI Accelerate

AI Accelerate is a curated self-learning track designed to establish a robust foundation in the rapidly evolving landscape of Artificial Intelligence. This module transitions learners from the core mechanics of Generative AI to the critical ethical frameworks required for enterprise-grade deployment. By combining conceptual e-learning with rigorous knowledge validation, this track ensures that GenCs are prepared to build and implement AI solutions that are both innovative and responsible.

Course Details

Course Title	Activity Code	Level
Fundamentals of Generative AI	ELRNG01863	101-Basics
Generative AI Quick Assessment Quiz	ATHDW335105	101-Basics
Introduction to Responsible AI	CGZ-914en	Foundational
Responsible AI-Deep Dive	ELRNG02344	101-Basics

Stage 1 Assessment

Day 21 - Afternoon, 22 - Forenoon

Stage 1 Assessment

- These parts of the days will be dedicated to the Stage 1 Assessment.

Assessment Overview

- **Assessment Type:** Skill-Based Assessment (SBA)
- **Duration:** 90 minutes
- **Number of Questions:** [C# - 2, SQL Server- 2, Web UI - 12]
- **Modules Covered:**
    - C#
    - SQL Server
    - Web UI (Multiple Choice Questions)
- **Maximum Attempts:** 3

Day 22 - Afternoon

- This part of the day has been included to adjust the Behavioral Training duration.

Stage 2 - Application Development and Maintenance Practices + Integrated Development Project (IDP)

Overview

Stage 2 focuses on Application Development and Maintenance Practices essential for the development and maintenance of various software applications. We offer a unique learning experience to learners by providing diverse learning content and methodologies based on adult learning principles.

As part of Stage 2 of your training, the following skills will be covered.

- Design Patterns and Principles, UML Basics
- Entity Framework Core 8.0
- ASP.NET Core 8.0
- Building RESTful APIs with ASP.NET Core 8.0
- Microservices Architecture
- Application Debugging using Visual Studio Debugger
- NUnit & Moq
- Cognizant Flowsource 101 foundation program [101-BASICS] (ELRNG02089)
- ITIL
- Jira, ServiceNow, Azure Boards
- Windows Service
- Python 3
- Cloud and DevOps Essentials
- GitHub Copilot

How and From Where to Learn?

- Udemy learnings are recommended in the Platform to understand the fundamental concepts. In addition to this, you can also learn from any other sources as they are mentioned in this handbook.

Stage 2 -> Course 1 -> Design Patterns and Principles, UML Basics

Course Overview

Course 1 of **Stage 2** will focus on **Design Principles and Patterns, UML Basics** and their practical implementations in various application development and maintenance scenarios.

Design Principles provide high level guidelines to design better software applications. They do not provide implementation guidelines and are not bound to any programming language. The SOLID (SRP, OCP, LSP, ISP, DIP) principles are one of the most popular sets of design principles.

Design Pattern provides low-level solutions related to implementation, of commonly occurring object-oriented problems. In other words, design pattern suggests a specific implementation for the specific object-oriented programming problem.

Design patterns are tested by others and are safe to follow, e.g., Gang of Four patterns: Abstract Factory, Factory, Singleton, Command, etc.

UML, or Unified Modeling Language, is a standardized way of visually representing a system's design in software engineering. It uses diagrams to show different parts of a system, like its structure or how it behaves. These diagrams help understand and communicate about the system's design throughout the software development process. UML includes various types of diagrams, each showing a different aspect of the system.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand and apply the Gang of Four (GoF) Design Patterns to solve common design problems in software development.
- Apply various software design principles, such as SOLID and DRY to enhance the quality and maintainability of software systems.
- Explain the basics of Unified Modeling Language (UML) and its importance in software development.
- Create and interpret different types of UML diagrams, including class diagrams, sequence diagrams, and activity diagrams.
- Identify and utilize the essential elements of UML diagrams, such as classes, objects, relationships, and behaviors, to model software systems effectively.

Day 23

Design Patterns and Principles, UML Basics

Key Topics: GoF Design Patterns

Continuous Learning: Technical Enablement

Design Patterns in C# and .NET

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 2:** Builder
  - **Section 3:** Factories
  - **Section 5:** Singleton
  - **Section 6:** Adapter
  - **Section 7:** Bridge
  - **Section 10:** Façade
  - **Section 13:** Chain of Responsibilities
  - **Section 14:** Command
  - **Section 20:** Observer
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Design Microservices Architecture with Patterns & Principles

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 2:** Monolithic Architecture
  - **Section 4:** Layered (N-Layer) Architecture
  - **Section 5:** Service-Oriented Architecture (SOA)
  - **Section 6:** Microservices Architecture
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- UA Highschool - Singleton Pattern
- GadgetHub - Factory Pattern
- SecureDocs - Proxy Pattern
- FoodiesDelight - Command Pattern

Day 24

Design Patterns and Principles, UML Basics

Key Topics: Different Types of Software Design Principles

Continuous Learning: Technical Enablement

Design Patterns in C# and .NET

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** The SOLID Design Principles
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Please find the hyperlink to an article on DRY principles in C# on the internet below and ensure you study its contents thoroughly.

- DRY Principles in C#

Day 25, 26 - Forenoon

Design Patterns and Principles, UML Basics

Key Topics: UML Diagram Types, UML Diagram Elements

Continuous Learning: Technical Enablement

The Complete UML Course: Learn to Design UML Diagrams

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Please find the hyperlink to an article on DRY principles in C# on the internet below and ensure you study its contents thoroughly.

- UML Diagram Elements

Stage 2 -> Course 2 -> Entity Framework Core 8.0

Course Overview

Course 2 of **Stage 2** provides a comprehensive introduction to **Entity Framework Core 8** (EF Core 8) and its seamless integration with .NET 8. Designed for aspiring developers, the course focuses on building robust and efficient data access layers for modern applications. Students will gain hands-on experience with creating database models, performing CRUD operations, executing LINQ queries, managing database migrations, and handling data relationships. The course also covers advanced topics such as performance optimization and best practices, ensuring that learners can develop scalable and maintainable applications with confidence.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the integration of EF Core 8 with .NET 8 and set up EF Core in a project.
- Design and implement a simple database model using EF Core 8.
- Perform basic CRUD (Create, Read, Update, Delete) operations with EF Core 8.
- Construct and execute LINQ queries for data retrieval and manipulation.
- Apply EF Core migrations to manage and update database schemas effectively.
- Handle data relationships and loading strategies, such as eager, lazy, and explicit loading.
- Optimize EF Core performance using best practices and advanced techniques.
- Build scalable and maintainable data access layers for modern software applications.

Day 26 - Afternoon

Entity Framework Core 8.0

Key Topics: Overview of EF Core 8 and .NET 8 Integration, Setting up EF Core in a .NET 8 Project

Continuous Learning: Technical Enablement

Entity Framework Core - A Full Tour

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Introduction
    - **Section 2:** Environment Setup
    - **Section 3:** Getting Started with Entity Framework Core
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Go through the below learning.

- Create a Model with Database Table in .NET 8 using EF Core

Day 27

Entity Framework Core 8.0

Key Topics: Creating a Simple Database Model, Performing Basic CRUD Operations, LINQ Queries in EF Core 8

Continuous Learning: Technical Enablement

Entity Framework Core - A Full Tour

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 4:** Using Entity Framework Core to Query a Database
  - **Section 5:** Using Entity Framework Core to Manipulate Data
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Go through the below learning.

- A Comprehensive Guide to Entity Framework Core in .NET 8
- CRUD Operations in Entity Framework Core

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Car Repository - Insert
- Car Repository - Eager Loading
- Car Repository - Lazy Loading

Day 28, 29 - Forenoon

Entity Framework Core 8.0

Key Topics: EF Core Migrations and Database Updates, Handling Relationships and Data Loading, Performance Optimizations and Best Practices

Continuous Learning: Technical Enablement

Entity Framework Core - A Full Tour

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 6:** Handling Database Changes and Migrations
  - **Section 7:** Interacting With Related Records
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Go through the below learning.

- EF Core Performance Optimisations

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on Entity Framework Core.

- Entity Framework

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac for checking your skill level on Entity Framework Core. You have to secure 70% in order to clear this challenge.

Do not copy paste the solution. Write the query yourself.

- Assess-Type-1: Code Challenge - Entity Framework

Stage 2 -> Course 3 -> ASP.NET Core 8.0

Course Overview

Course 3 of **Stage 2** provides a comprehensive introduction to **ASP.NET Core 8**, a modern, open-source, cross-platform framework for building web applications. Designed for aspiring developers, the course emphasizes practical skills to design, develop, and deploy dynamic web applications using the latest features of ASP.NET Core. Participants will gain a solid understanding of the MVC (Model-View-Controller) architecture, explore robust techniques for handling user input and state management, and learn to integrate Entity Framework (EF) Core 8 for seamless data operations. By the end of the course, learners will be equipped to create scalable, maintainable, and deployable web applications using industry best practices.

Learning Objectives

After completing this course, GenCs will be able to:

- Describe the key features and benefits of ASP.NET Core 8.
- Set up and configure a new ASP.NET Core project.
- Explain the principles of the Model-View-Controller (MVC) design pattern.
- Develop and integrate Models, Controllers, and Views effectively.
- Create and manage forms to collect and process user input.
- Utilize routing to enable intuitive navigation and URL structures.
- Explore techniques for session and application state management.
- Implement secure state management solutions.
- Configure and connect ASP.NET Core applications with databases using EF Core 8.
- Perform CRUD (Create, Read, Update, Delete) operations with EF Core.
- Implement Dependency Injection (DI) for modular and testable code.
- Understand the process of publishing and deploying ASP.NET Core applications to various hosting environments.
- Apply the skills and concepts learned to design and develop a fully functional web

application, ready for deployment.

Day 29 - Afternoon

ASP.NET Core 8.0

Key Topics: Introduction to ASP.NET Core 8, Creating a New ASP.NET Core Project, Understanding MVC Architecture

Continuous Learning: Technical Enablement

The complete ASP.NET Core 9 course for busy developers

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Building your first ASP.NET Core application
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Interrogation Panel - Data Transferring
- Interrogation Panel - Component
- Interrogation Panel - Razor Page

Day 30

ASP.NET Core 8.0

Key Topics: Working with Models, Building Controllers, Designing Views in ASP.NET Core

Continuous Learning: Technical Enablement

- Get started with ASP.NET Core MVC
- Add a controller to an ASP.NET Core MVC app
- Add a view to an ASP.NET Core MVC app
- Add a model to an ASP.NET Core MVC app

Day 31

ASP.NET Core 8.0

Key Topics: Routing in ASP.NET Core, Working with Forms and User Input

Continuous Learning: Technical Enablement

- Routing in ASP.NET Core
- ASP.NET Core Blazor forms overview

Asp.Net Core 10 (.NET 10) | True Ultimate Guide

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 5:** Routing [MVC and Web API]
  - **Section 17:** Tag Helpers [MVC]
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 32, 33

ASP.NET Core 8.0

Key Topics: State Management, Integrating EF Core 8 with ASP.NET Core

Continuous Learning: Technical Enablement

- Session and state management in ASP.NET Core

Continuous Learning: Technical Enablement

Asp.Net Core 10 (.NET 10) | True Ultimate Guide

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 17:** Tag Helpers [MVC]
  - **Section 18:** Entity Framework Core [MVC and Web API]
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Interrogation Panel - DB Initialize

Day 34, 35 - Forenoon

ASP.NET Core 8.0

Key Topics: Dependency Injection (DI) in ASP.NET Core, Publishing and Deployment

Continuous Learning: Technical Enablement

- Dependency injection in ASP.NET Core
- Host and deploy ASP.NET Core

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac for checking your skill level on ASP.NET Core. You have to secure 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1 Code Challenge - ASP.Net Core, MVC with Entity Framework

Interim Evaluation

Day 35 - Afternoon, 36, 37 - Forenoon

Interim Evaluation (Project + Technical)

- Interim evaluation will be conducted on these days, and the mode will be a video interview on the Tekstac platform.

Course Overview

Course 4 of **Stage 2** provides an in-depth understanding of **Web APIs and ASP.NET Core**, enabling learners to design, develop, and secure robust API solutions. Starting with the foundational concepts of Web APIs and ASP.NET Core, the course delves into building RESTful APIs, exploring advanced API features, and integrating SOAP services. Additionally, learners will gain expertise in API security, exception handling, and best practices for API documentation and testing. By the end of the course, participants will be equipped to create scalable, secure, and well-documented APIs tailored to real-world applications.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the fundamental principles of Web APIs and the ASP.NET Core framework.
- Design and implement RESTful APIs using ASP.NET Core, adhering to industry standards.
- Explore advanced API features, including pagination, versioning, and filtering.
- Integrate and consume SOAP services in .NET applications.
- Ensure API security through authentication, authorization, and secure communication practices.
- Effectively handle exceptions and implement error-handling mechanisms in APIs.
- Create comprehensive API documentation using tools like Swagger/OpenAPI.
- Conduct rigorous testing of APIs to ensure reliability and performance.

Day 37 - Afternoon

Building RESTful APIs with ASP.NET Core 8.0

Key Topics: Introduction to Web APIs and ASP.NET Core, Building RESTful APIs with ASP.NET Core

Continuous Learning: Technical Enablement

ASP.NET CORE WEB API | The Complete Guide

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Before You Get Started
  - **Section 2:** Introduction to Web API
  - **Section 3:** Building Your First ASP.NET Core API
  - **Section 5:** Controller Action Return Types
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Go through the below learning.

- Implement Entity Framework A Code First Approach in .Net 8 API

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Online Bookstore API - Insert

Day 38

Building RESTful APIs with ASP.NET Core 8.0

Key Topics: Advanced API Features

Continuous Learning: Technical Enablement

Build ASP.NET Core Web API - Scratch To Finish (.NET8 API)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 8:** Securing our ASP.NET Core API - Authentication and Authorization - JWT Tokens
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Go through the below learning.

- Routing in ASP.NET Core Web API
- Filters in ASP.NET Core Web API

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Online Bookstore API – Retrieve
- Online Bookstore API - Modify
- Online Bookstore API – Remove

GitHub Copilot License Activation

The GitHub Copilot license activation is a centralized, batch owner-led process designed to streamline the onboarding of AI-assisted coding tools for the entire cohort. Rather than individual requests, the Batch Owner initiates a consolidated request for all cohort members through the designated PoC once the program reaches the appropriate technical phase. This ensures that enterprise license slots are allocated efficiently and that all GenCs receive synchronized access to Copilot features within their IDEs, supporting a unified learning experience as they transition into AI-driven development.

Day 39, 40

Building RESTful APIs with ASP.NET Core 8.0

Key Topics: Consuming and Creating SOAP Services, API Security and Exception Handling

Continuous Learning: Technical Enablement

- SOAP Web Service in .NET Core

Continuous Learning: Technical Enablement

Build ASP.NET Core Web API - Scratch To Finish (.NET8 API)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 10:** Advanced Functionality in ASP.NET Core Web APIs
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 41

Building RESTful APIs with ASP.NET Core 8.0

Key Topics: API Documentation and Testing

Continuous Learning: Technical Enablement

Asp.Net Core 8 (.NET 8) | True Ultimate Guide

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 27:** Swagger / Open API
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac for checking your skill level on ASP.NET Core Web API. You have to secure 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Web API.Net

Stage 2 -> Course 5 -> Microservices Architecture

Course Overview

Course 5 of **Stage 2** will be focusing on the fundamental aspects of **Microservices** and their core features**.**

Software architecture using microservices is an architectural style where an application is composed of small, independent services that are built around specific business capabilities and communicate with each other through well-defined APIs. Each microservice is designed to be modular, highly maintainable, and independently deployable, allowing for flexibility and scalability in large and complex applications.

Learning Objectives

After completing this course, GenCs will be able to:

- Differentiate between Monolithic and Microservices architectures and articulate the trade-offs regarding scalability, fault isolation, and deployment independence.
- Apply Domain-Driven Design (DDD) basics to define clear service boundaries and responsibilities for modular services like Product, Order, and User.
- Build production-ready microservices using ASP.NET Core Web API, leveraging Dependency Injection, configuration management, and Entity Framework Core for data persistence.
- Implement Synchronous communication using HttpClientFactory and Ocelot API Gateways, and manage service resolution through discovery tools like Consul or Eureka.
- Design Asynchronous messaging workflows using RabbitMQ or Azure Service Bus to facilitate decoupled, event-driven interactions between services.
- Package microservices using Docker and explain the role of Kubernetes in orchestrating independent service scaling and lifecycle management.
- Resilience Engineering: Integrate Health Checks and stability patterns—such as Retry and Circuit Breaker—to ensure system reliability in a distributed environment.

Day 42

Microservices Architecture

Key Topics: Fundamentals of Microservices Architecture, Designing Microservices with ASP.NET Core

Continuous Learning: Technical Enablement

.NET Core Microservices - The Complete Guide [2025]

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Introduction
  - **Section 2:** Section 2 Coupon API - Getting Started
  -
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learning References (Source: https://learn.microsoft.com/)

- Build your first microservice with .NET

Day 43

Microservices Architecture

Key Topics: Inter-Service Communication

Continuous Learning: Technical Enablement

.NET Core Microservices - The Complete Guide [2025]

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 10:** Section10 Service Bus
  - **Section 17:** Gateway
  - **Section 19:** RabbitMQ Implementation
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 44

Microservices Architecture

Key Topics: Microservices Deployment and Scalability

Continuous Learning: Technical Enablement

.NET Core Microservices - The Complete Guide [2025]

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 18:** Azure Deployment

- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learning References (Source: https://learn.microsoft.com/)

- Deploy a .NET microservice to Kubernetes

Day 45 - Forenoon

- This part of the day has been included in adjusting the Behavioral Training duration.

Stage 2 -> Course 6 -> Application Debugging using Visual Studio Debugger

Course Overview

Course 6 of **Stage 2** will be focusing on **Application Debugging using Visual Studio Debugger** which can help you navigate through code to inspect the state of an app and show its execution flow.

Debugging is the process of detecting and removing of existing and potential errors (also called as 'bugs') in a software code that can cause it to behave unexpectedly or crash. To prevent incorrect operation of a software or system, debugging is used to find and resolve bugs or defects. When various subsystems or modules are tightly coupled, debugging becomes harder as any change in one module may cause more bugs to appear in another. Sometimes it takes more time to debug a program than to code it.

Learning Objectives

After completing this course, GenCs will be able to

- Explain what Debugging is and why do we need it
- Use Visual Studio Debugger that helps in navigating through code to inspect the state of an app and show its execution flow.
- Employ various debugging techniques during application development and maintenance.

Day 45 - Afternoon

Application Debugging using Visual Studio Debugger

Key Topics: Introduction to Debugging, Visual Studio Debugger, Navigate through code by using the Visual Studio debugger

Continuous Learning: Technical Enablement

Visual Studio Mastery with C# - Double Your Productivity

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 6:** Debugging Tools in Visual Studio
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learn about Application Debugging from the following

- Debugging in Visual Studio
- Debugging in C# - Finding and Fixing Problems in Your Application

Stage 2 -> Course 7 -> NUnit & Moq

Course Overview

Course 7 of **Stage 2** will be focusing on **Unit Testing and Mocking Frameworks such as NUnit, Moq** and their implementation during Test Driven Development.

Unit Testing is a software testing method by which individual units of source code—sets of one or more computer program modules together with associated control data, usage procedures, and operating procedures are tested to determine whether they are fit for use.

NUnit is an open-source unit testing framework for the .NET Framework and Mono. It serves the same purpose as JUnit does in the Java world and is one of many programs in the xUnit family.

Moq is a mocking framework built to facilitate the testing of components with dependencies.

Learning Objectives

After completing this course, GenCs will be able to

- Explain the Test Pyramid and identify the role of automated unit tests in reducing regression risks and improving software quality.
- Implement the Test-Driven Development (TDD) workflow using NUnit within Visual Studio.
- Apply industry-standard naming conventions and organizational structures to create high-quality, readable test suites.
- Execute unit tests for diverse data structures, including strings, collections, and arrays, while effectively handling exception testing and void methods.
- Refactor legacy or tightly coupled code into a testable design by applying Dependency Injection (DI) patterns (Constructor, Property, and Method injection).
- Use the Moq framework to create mock objects and stubs, enabling the isolation of the "System Under Test" (SUT) from external dependencies like databases or APIs.
- Contrast State-based testing with Interaction testing to verify both final outcomes and the behavior between collaborating objects.

- Utilize Code Coverage tools to identify untested logical paths and ensure the reliability of the codebase.

Day 46

NUnit & Moq

Key Topics: Getting Started, Fundamentals of Unit Testing, Core Unit Testing Techniques

Continuous Learning: Technical Enablement

Unit Testing for C# Developers

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Getting Started
    - **Section 2:** Fundamentals of Unit Testing
    - **Section 3:** Core Unit Testing Techniques
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Test Case - Validations On String Concatenation
- Test Case - Validations On Vowel Checker
- Test Case - Validations On Array Search
- Test Case - Validations on Array Multiplication
- Test Case - Validations on List

Day 47

NUnit & Moq

Key Topics: Moq-Breaking External Dependencies

Continuous Learning: Technical Enablement

Unit Testing for C# Developers

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 5:** Breaking External Dependencies
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 48 - Forenoon

Cognizant Flowsource 101 Foundation Program

Cognizant Flowsource is an AI-native, full-stack modern engineering platform that uses generative and agentic AI to speed up the entire Software Development Lifecycle (SDLC), aiming to boost developer productivity, improve code quality, and accelerate time-to-market. It provides integrated tools, prompts, and domain-trained agents that assist product managers, architects, and engineers with tasks from ideation to deployment, offering transparency and governance with built-in security and quality checks. Flowsource acts as an extensible platform, allowing integration with third-party tools like Jira, and helps teams shift from non-value-add work to strategic focus.

Key Features & Capabilities:

- **AI-Powered SDLC:** Integrates generative AI and automation across all development stages, from requirements to testing and deployment.
- **Unified Experience:** Offers a single platform for designers, engineers, and product owners, reducing fragmentation.
- **Productivity & Quality:** Drives significant productivity gains (often 25-40%) while ensuring high-quality, well-architected code through embedded practices and quality checks.
- **Agentic AI:** Uses domain-trained agents and prompt libraries to automate complex tasks, like generating user stories from documents, reducing weeks of work to minutes.
- **Transparency & Governance:** Provides insights into engineering throughput and quality, with built-in security and responsible AI principles.
- **Extensible:** Allows for third-party plugins and integration with existing tools (e.g., Jira) for customized workflows.

Benefits:

- **Faster Innovation:** Accelerates the delivery of new products and features, improving time-to-market.
- **Reduced Technical Debt:** Helps teams tackle legacy issues and complex code more efficiently.
- **Focus on Value:** Frees up engineering talent from repetitive tasks to focus on strategic, high-impact work.
- **Business Impact:** Delivers tangible results, such as significant cost savings and increased developer efficiency, as seen with some clients.

Register and complete the following E-Learning from CLearn.

- Cognizant Flowsource 101 foundation program [101-BASICS]

Course Overview

Course 8 of **Stage 2** provides a comprehensive understanding of the ITIL 4 framework, its evolution from ITIL v3, and its application in modern IT service management. The program is designed to equip GenCs with the foundational knowledge of service management principles, the Service Value System (SVS), and the Service Value Chain (SVC). Participants will explore the roles of organizations, people, technology, and partners in creating and delivering value through service management practices.

Additionally, this course delves into the guiding principles of ITIL 4, offering practical insights into their application. GenCs will gain expertise in service management practices, continual improvement, and the integration of ITIL's best practices into organizational processes.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain the key concepts and principles of ITIL 4.
- Describe the evolution from ITIL v3 to ITIL 4 and its relevance to modern service management.
- Understand the definition, scope, and principles of service management.
- Describe the components and significance of the ITIL 4 Service Value System (SVS).
- Identify the roles of organizations, people, information, technology, partners, and suppliers in the SVS.
- Illustrate value streams and processes within the service management lifecycle.
- Explain the activities and interconnections within the SVC, including planning, improving, engaging, designing, obtaining/building, and delivering/supporting services.
- Understand and apply general management practices such as service level management, service catalog management, availability management, and more.
- Implement ITIL practices like monitoring, deployment, release, and service request management.
- Explain the guiding principles of ITIL 4 and their role in service management.
- Apply principles such as focusing on value, iterative progression, collaboration, simplicity, and optimization in service delivery.
- Understand the continual improvement model and its importance in ITIL 4.
- Apply measurement and reporting techniques to drive improvement initiatives.

Day 48 - Afternoon, 49 - Forenoon

ITIL Framework

Key Topics: Introduction to ITIL 4, Service Management, Four Dimensions of Service Management, ITIL 4 Service Value System (SVS), ITIL 4 Service Value Chain (SVC), ITIL 4 Practices, ITIL 4 Guiding Principles, ITIL 4 Continual Improvement

Continuous Learning: Technical Enablement

Introduction to Service Management with ITIL 4

- Go through ALL sections of this course to understand ITIL 4.
- Take up the Practice Exam given as part of this course to check your understanding about ITIL 4.

Quiz - Mandatory

Take up the following quiz to assess your knowledge on the ITIL Framework.

- ITIL Quiz

Stage 2 -> Course 9 -> Windows Service

Course Overview

Course 9 of Stage 2 will be focusing on **Windows Service** which is a core component of the Microsoft Windows operating system and enables the creation and management of long-running processes

Unlike regular software that is launched by the end user and only runs when the user is logged on, Windows Services can start without user intervention and may continue to run long after the user has logged off. The services run in the background and will usually kick in when the machine is booted. Developers can create Services by creating applications that are installed as a Service, an option ideal for use on servers when long-running functionality is needed without interference with other users on the same system.

The services manage a wide variety of functions including network connections, speaker sound, data backup, user credentials and display colors. Windows Services perform a similar function as UNIX daemons.

Learning Objectives

After completing this course, GenCs will be able to

- Explain what is a Windows Service, why we need it and what are the differences between Windows Services and Regular Applications.
- Develop and manage a Windows Service.

Day 49 - Afternoon

Windows Service

Key Topics: Introduction, Developing and Managing a Windows Service

Continuous Learning: Technical Enablement

Please find the hyperlink to an article on Windows Service on the internet below and ensure you study its contents thoroughly.

- Windows Service
- TaskScheduler Class
- Quartz API

Stage 2 -> Course 10 -> Python 3

Course Overview

Course 10 of **Stage 2** This course provides a comprehensive introduction to **Python**, a versatile and widely used programming language. Designed for beginners and intermediate learners, the curriculum covers essential programming concepts, Python-specific features, and advanced techniques. GenCs will explore Python's foundational constructs, control flow, data structures, object-oriented programming (OOP), file handling, exception management, and modular programming. Through practical exercises and hands-on coding, GenCs will develop the skills needed to write efficient, readable, and scalable Python programs.

Learning Objectives

After completing this course, GenCs will be able to:

- Apply Python syntax to build foundational scripts, utilizing appropriate data types, operators, and control flow statements (if/else, loops) to solve algorithmic problems.
- Evaluate and implement Python Collections (Lists, Tuples, and Dictionaries) to manage, sort, and manipulate complex datasets efficiently.
- Design modular code using Functions, demonstrating proficiency in parameter passing, variable-length arguments, and the effective use of Lambda functions and variable scoping.
- Architect scalable software using Object-Oriented Programming principles, including class construction, inheritance, method overriding, and the use of private properties for encapsulation.
- Implement robust Exception Handling and assertions to manage runtime errors and ensure application stability during unexpected inputs or system failures.
- Execute file-system operations and organize codebases into maintainable Modules and Packages, utilizing namespaces to avoid naming conflicts.
- Develop dynamic web applications using Flask or Django, implementing custom routing, template rendering, and middleware.
- Build and secure RESTful APIs using JSON handling, HTTP status codes, and industry-standard authentication protocols like JWT and RBAC.
- Integrate relational and non-relational databases into web applications using Object-Relational Mapping (ORM) to perform complex CRUD operations and data modeling.

Day 50

Python 3

Key Topics: Basics, Variables and Types, Program Flow, Functions

Continuous Learning: Technical Enablement

Complete Python Programming Masterclass Beginner to Advanced

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 2:** Getting Setup with Python
  - **Section 3:** Variables and Types
  - **Section 8:** Python Program Flow
  - **Section 13:** Python Functions
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Welcome Message
- Virtual Assistant
- Hospital Billing
- Stationary Shop
- Theme Park Pricing
- Cash Exchange
- House Rent Calculator
- Airline Reservation System
- Voting System
- Fitness Tracker Analyzer

Day 51

Python 3

Key Topics: Python Operators, Collections, Object Oriented Programming (OOP)

Continuous Learning: Technical Enablement

Complete Python Programming Masterclass Beginner to Advanced

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 4:** Python Operators

        - **Section 13:** Python Collections
        - **Section 14:** Python Object Oriented Programming (OOP)
    - Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Non-Working Doctors
- Product Sales
- Participant Registration and Preference Analysis
- Student Electives
- Sort the Student Details Dictionary
- Generate Customer Id
- Online Shopping Cart System
- Pattern Generator
- Vehicle Premium
- Classic Curve Members
- Area calculation - Method Overrding
- Tournament - Method overloading
- Complex Addition

Day 52, 53, 54 - Forenoon

Python 3

Key Topics: File I/O, Exception Handling, Modules, Web Frameworks, RESTful API Development, Database Integration, Authentication & Authorization

Continuous Learning: Technical Enablement

Complete Python Programming Masterclass Beginner to Advanced

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 10:** Working With Files
    - **Section 15:** Handling Errors in Python
    - **Section 7:** Python Modules
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Complete 2026 Python Bootcamp: Learn Python from Scratch

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 13:** Building Web Applications using Flask

- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learning References

- Building a Simple API with Django REST Framework
- Flask SQLAlchemy Tutorial for Database
- Django ORM - Inserting, Updating & Deleting Data
- Flask API Authentication with JSON Web Tokens

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Basket Ball Player Analysis
- Percentage of marks - Lamda Functions
- Employee Appraisal
- Product Information
- Filter the Countries Sachin has Played Against - CSV file
- Filter Customers - JSON File
- Data Encryption
- Password Update
- Lucky Number – Generator
- Project Allocation – Iterator
- Summer Camp - Decorator

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac to check your skill level in Python 3. You need to score 70% or higher to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Python

Stage 2 -> Course 11 -> Cloud and DevOps Essentials

Course Overview

Course 11 of **Stage 2** provides an end-to-end journey through **cloud-native architecture** and the modern **DevOps lifecycle**, transitioning from foundational cloud principles like elasticity and service models to advanced enterprise implementation on Microsoft Azure. Learners will master the deployment of core compute, storage, and networking resources while simultaneously **building automated CI/CD pipelines using Azure DevOps** to streamline software delivery. By combining

Infrastructure as Code (IaC), sophisticated deployment strategies such as Canary and Blue-Green releases, and proactive monitoring via Application Insights, this module empowers GenCs to build, deploy, and maintain highly available, production-ready applications in a cloud-first environment.

Learning Objectives

After completing this course, GenCs will be able to

- Distinguish between IaaS, PaaS, and SaaS service models and select appropriate deployment models (Public, Private, Hybrid) based on business requirements.
- Navigate the Azure Global Infrastructure and utilize the Azure Portal, CLI, and Resource Manager (ARM) to deploy and manage cloud resources.
- Provision and configure Virtual Machines, scalable App Services, and serverless Azure Functions, while managing persistent data using Blob, File, and Table storage.
- Design secure network architectures using Virtual Networks (VNETs) and Subnets, and enforce security policies through Network Security Groups (NSGs) and Role-Based Access Control (RBAC) via Azure AD.
- Apply DevOps principles using Azure Repos and Artifacts to manage source control and package dependencies within a professional development lifecycle.
- Construct automated CI/CD pipelines using both YAML and Classic editors to enable continuous integration and reliable delivery of software to Azure environments.
- Execute sophisticated release patterns, including Blue-Green, Canary, and Rolling deployments, using App Service deployment slots for zero-downtime releases. [Image comparing Blue-Green and Canary deployment strategies]
- Describe the benefits of automated provisioning and create basic templates using ARM or Terraform to maintain consistent environments.
- Implement proactive monitoring and diagnostic logging using Azure Monitor and Application Insights to track application performance and health.

Day 54 - Afternoon, 55

Cloud and DevOps Essentials

Key Topics: Cloud Fundamentals, Azure Essentials, Compute & Storage Services, Networking & Security

Continuous Learning: Technical Enablement

AZ-900 Microsoft Azure Fundamentals + FULL Practice Exam!

- Learn the sections listed below in this Udemy course and complete the corresponding hands-on coding given below.
    - **ALL Sections**
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 56, 57 - Forenoon

Cloud and DevOps Essentials

Key Topics: CI/CD with Azure DevOps, Deployment Strategies, Monitoring

Continuous Learning: Technical Enablement

Azure DevOps Fundamentals for Beginners

- Learn the sections listed below in this Udemy course and complete the corresponding hands-on coding given below.
    - **ALL Sections**
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Integrated Capability Test (ICT) (For Practice Only)

Attempt the Integrated Capability Test (ICT) available on the Learning Path at Tekstac to assess your skill level in Stage 2, particularly focusing on web application development skills. You must achieve a score of 70% or higher to pass this practice test.

Do not copy paste the code. Write the code yourself.

- Dotnet Assess-Type-2: Integrated Capability Test (ICT)

Day 57 - Afternoon, 58, 59 - Forenoon

GitHub CoPilot

This session begins with an expert-led Masterclass, where learners are introduced to prompt engineering, code generation patterns, and the ethical guardrails of using AI in development. This is followed by intensive Hands-on Exercises where GenCs apply Copilot to diverse real-world use cases, including legacy code refactoring, unit test generation, and complex logic implementation. By mastering these AI capabilities during the training phase, GenCs ensure they are project-ready and capable of delivering high-velocity, high-quality code before their final release to the Business Unit.

Final Evaluation (Project + Technical)

Day 59 - Afternoon, 60, 61, 62 - Forenoon

Final Evaluation (Project + Technical)

- Final Evaluation will be conducted on these days, and the mode will be a video interview on the Tekstac platform.

Final HackerRank Assessment

Day 62 - Afternoon, 63 - Forenoon

HackerRank Assessment (C#, SQL)

- The final assessment will be conducted on this day.

How to learn each day?

Each day has a set of learning objectives. These learning objectives can be met by going through the Udemy courses and by completing the hands-on exercises mentioned in the daily plan.

The below strategies will help you decide the learning approach.

Learning Strategy & Approach

Find below few imaginary profiles. For each of these profiles we have defined a recommended learning approach. This is not an exhaustive list. The approaches below might help invent a new way of learning.

Profile #1

Harry Reacher

Engineering Discipline: Electronics

Skills: Python, Ruby on Rails, nginx

Project: Mining Crime Data to get Route Cause Insights

Learning Approach to Programming Languages: I do not want to waste my time learning. I am more practice oriented. I want to work on the problem immediately

What will work for me?

- Directly complete hands-on exercises
- Refer Internet or Udemy Courses
- If hands on are implemented early, clarify your friends' questions and troubleshoot their issues

Profile #2

Olivia Richards

Engineering Discipline: Computer Science

Skills: Java, C, C++

Project: Library Management System

Learning Approach to Programming Languages: I have interest, but I don't know where to start.

What will work for me?

- Go through the recommended Udemy Course
- Try completing the hands-on exercises
- Get your clarifications solved with help from Tech SME
- Get help from other learners in your batch whom had already completed

Profile #3

Greg Anderson

Engineering Discipline: Civil

Skills: C

Project: Fiber reinforced concrete

Learning Approach to Programming Languages: I am scared of programming languages. I haven't got my hands dirty with coding

What will work for me?

- Go through the recommended Udemy Course
- Implement the coding along with the author of the Udemy Course
- Try completing the hands-on exercises
- Clarify queries with SME
- Troubleshoot programming issues with help from SME or learner from your classroom whom had already completed

FAQs

1. **Who can participate in this program?**

   Ans: Students who have enrolled for Full Internship Program (or) the Cognizant on-boarded GenCs can participate in this program.

2. **Is there any pre-learning I should do?**

   Ans: No. This program is open to all students from any academic discipline.

3. **What is Code Challenge?**

   Ans: A problem statement will be provided to you and you need to solve it using a single skill.

4. **What is Integrated Capability Test (ICT)?**

   Ans: A case study problem statement will be provided to you that you may need to solve using the combination of skills learned in the given stage.

5. **How many attempts are provided for the Coding challenge and ICTs? Is it open all the time for practice?**

   Ans: The coding challenges and ICTs are open from day 1, and a maximum of 3 attempts will be provided.

6. **What are the entry criteria for Stage 1 Assessment?**

   Ans: The eligibility criterion for the Stage 1 Assessment is 100% hands-on completion, along with an attempt and submission in both CC and ICT.

7. **What skills are covered in the Stage 1 Assessment?**

   Ans: This assessment covers foundational skills such as C#, SQL, and Web UI technologies. You will have a maximum of three attempts to pass, with a minimum required score of 70%.

8. **What if I fail in the Interim evaluation?**

   Ans: Your coach will notify your performance in the Interim evaluation. However, you can continue with the learning.

9. **How many chances will I get in the Final evaluation?**

   Ans: You'll get 2 chances in the Final evaluation which covers ALL the skills in the learning journey.

10. **Will we be provided with Projects to work on?**

    Ans: Yes, we will provide the requirement specifications at the beginning phase of your Stage 2 training. You will need to implement them by forming a PoD (Pod of Developers). The implementation will be reviewed during both the interim evaluation and the final evaluation.

11. **Whom do I reach out in case of any queries?**

    Ans: Coach is your point of contact.